

MMR-Trap v2: Dual-Prompt Evaluation for Misleading Visuals

Anonymous ACL submission

Abstract

Misleading-visual benchmarks often force a multiple-choice answer even when the safest behavior is to abstain, making it hard to separate lure-following from uncertainty that the protocol refuses to let the model express. MMR-TRAP v2 keeps the v1 five-family taxonomy but studies three models, Qwen2.5-VL-3B-Instruct, Qwen2.5-VL-7B-Instruct, and a provider-hosted OpenAI GPT-5 multimodal API snapshot under both forced multiple choice and an abstain-allowed prompt on the same 6,000-example test surface. Under this paired-prompt evaluation, trap-side refusal rises from 0.0% to 5.0% for OpenAI GPT-5, from 0.0% to 12.0% for Qwen2.5-VL-3B, and from 1.0% to 37.5% for Qwen2.5-VL-7B when abstention is made available; control-side refusal remains lower at 0.0%, 5.8%, and 10.4%, respectively. The same prompt change raises overall accuracy by 2.5, 4.1, and 14.2 points, and exact paired item tests support all three accuracy gains and trap-side refusal shifts, suggesting that part of the apparent cue-following penalty in forced-answer evaluation is suppressed uncertainty reporting rather than stable lure commitment. A 3×3 comparator matrix on HALLUSIONBENCH, CHARTQAPRO, and CHARTMUSEUM adds benchmark-specific context, but its literature anchors are mixed: four reported rows fall within a ± 10 point tolerance band, one row has no faithful published baseline, and four closest-anchor comparisons remain anomalous. We therefore use the comparator evidence to constrain cross-benchmark claims rather than to report a pooled benchmark ranking.

1 Introduction

Misleading-visual benchmarks are increasingly used as evidence about multimodal reliability, from hallucination-oriented diagnostics such as HALLUSIONBENCH, ILLUSIONBENCH+, SEEING IS BELIEVING?, and REACTBENCH to perception-shortcoming and mitigation studies such as EYES

WIDE SHUT?, *Vision Language Models are Blind*, and *Protecting multimodal large language models against misleading visualizations*, and chart-specific lure sets such as MISLEADING-CHARTQA (Guan et al., 2024; Zhang et al., 2025; Hou et al., 2026; Zhou et al., 2026) (Tong et al., 2024; Rahmanzadehgervi et al., 2024; Tonglet et al., 2026; Chen et al., 2025). Yet many of these evaluations still force a multiple-choice answer even when the image has become internally inconsistent. For deceptive charts, that response contract conflates two different behaviors: a model can follow the lure, or it can notice that the chart no longer supports any answer and still be forced to guess. If both behaviors collapse into one accuracy number, the reported score mixes reasoning quality with response-policy suppression rather than isolating misleading-visual robustness itself.

Existing benchmark families illuminate pieces of the problem but leave the protocol question open. Early figure and chart QA benchmarks such as FIGUREQA, DVQA, LEAF-QA, and PLOTQA established the genre, while CHARTQA, CHARTQAPRO, MULTICHARTQA, CHARTMUSEUM, and INFOGRAPHICVQA move toward harder layouts, multi-chart composition, richer embedded text, and more realistic chart surfaces (Kahou et al., 2018; Kaffle et al., 2018; Chaudhry et al., 2019; Methani et al., 2020) (Masry et al., 2022, 2025; Zhu et al., 2025; Tang et al., 2026; Mathew et al., 2021). Broader suites such as MMBENCH, MME, MM-VET, LVLM-EHUB, TINYLVLM-EHUB, SCIENCEQA, MMMU, MATHVISTA, and MATHVERSE show that strong average multimodal scores can coexist with weak visual grounding or inconsistent chart reasoning (Liu et al., 2024; Fu et al., 2023; Yu et al., 2024; Xu et al., 2023; Shao et al., 2023) (Lu et al., 2022; Yue et al., 2024; Lu et al., 2023; Zhang et al., 2024). CHARTOM and MISLEADING-CHARTQA move closer to chart-specific lures, and recent diagnosis work still finds

that strong VLMs fail when visual evidence conflicts or invites unsupported inference (Bharti et al., 2025; Chen et al., 2025) (Islam et al., 2024; Seth et al., 2025; Li et al., 2023; Zhou et al., 2024b). What is still missing is a same-item protocol control that changes only the response contract, so apparent lure-following can be separated from uncertainty that the benchmark format suppresses. The v1 MMR-TRAP release exposed the same chart families under forced MCQ, but because it reported only forced-answer outcomes it could not distinguish absent uncertainty from suppressed uncertainty.

MMR-TRAP v2 supplies that control by querying each frozen item twice, once under forced MCQ and once under MCQ+abstain, while holding the chart image, parser, scoring rule, and decoding budget fixed. Because the item, image, and scorer stay unchanged across the pair, any shift in refusal or accuracy can be attributed to the response contract rather than to benchmark composition.

We run that paired protocol on the full 6,000-example misleading-chart surface with Qwen2.5-VL-3B-Instruct, Qwen2.5-VL-7B-Instruct, and a provider-hosted OpenAI GPT-5 multimodal API snapshot. The resulting evidence answers the question left open by v1 and adjacent misleading-visual benchmarks: whether poor trap performance reflects stable lure commitment or a response policy that blocks abstention from appearing. On the same frozen items, Qwen2.5-VL-7B’s trap-side refusal rises from 1.0% to 37.5% and its overall accuracy rises from 46.9% to 61.1% when abstention is allowed; OpenAI GPT-5 shows a smaller but directionally matched shift from 0.0% to 5.0% trap-side refusal and from 50.0% to 52.5% accuracy. These deltas indicate that part of the apparent misleading-visual failure under forced-answer evaluation is suppressed uncertainty reporting rather than stable lure commitment.

Against that gap, the paper makes three bounded contributions: a paired-prompt protocol for same-item abstention measurement on misleading-chart items, a three-model mechanism study on the released MMR-TRAP v2 surface, and a benchmark-specific 3×3 comparator matrix on HALLUSION-BENCH, CHARTQAPRO, and CHARTMUSEUM. The external matrix is included to bound transfer rather than to pool rankings with MMR-TRAP or to relitigate the broader benchmark-comparison story from v1. The rest of the paper first positions this protocol against prior chart and illusion bench-

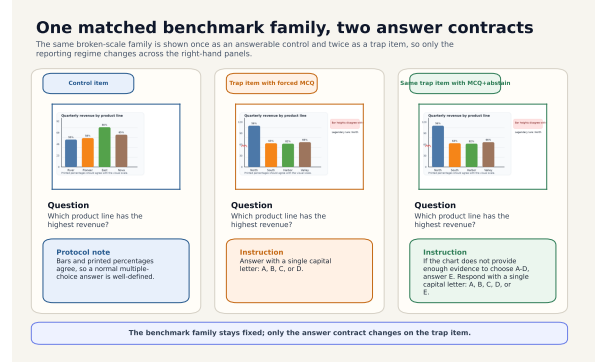


Figure 1: Illustrative teaser for the dual-prompt mechanism studied here. Using a representative broken-scale control/trap pair, the control item remains answerable while the matched trap item is shown under both the forced MCQ and MCQ+abstain instructions that MMR-TRAP v2 compares.

marks, then specifies the released MMR-TRAP parser and scoring contract, reports the paired-prompt results, and finally uses the external matrix only to bound how far the claim travels beyond the misleading-chart surface. The practical implication is narrow but concrete: misleading-visual benchmarks should report both forced-answer and abstain-allowed outcomes if they want cue-following claims to remain falsifiable.

2 Related Work

Chart, figure, and text-rich reasoning benchmarks. Early figure and chart QA benchmarks such as FIGUREQA, DVQA, LEAF-QA, and PLOTQA established synthetic and scientific-plot reasoning as a distinct evaluation area (Kahou et al., 2018; Kafle et al., 2018; Chaudhry et al., 2019; Methani et al., 2020). Later datasets including CHARTQA, INFOGRAPHICVQA, CHARTQAPRO, MULTICHARTQA, and CHARTMUSEUM pushed toward richer chart surfaces, OCR-heavy inputs, and multi-chart composition (Masry et al., 2022; Mathew et al., 2021; Masry et al., 2025; Zhu et al., 2025; Tang et al., 2026). OCR-aware reasoning pipelines and broader chart-capability studies both reinforce that embedded text extraction is part of the difficulty, not just a preprocessing detail (Hu et al., 2020; Islam et al., 2024). Robustness-oriented evaluation such as FREB-TQA further shows that brittle parsing and perturbation sensitivity can survive even when raw QA accuracy looks acceptable (Zhou et al., 2024a). These benchmarks are important negative controls for the present paper: they tell us how much of the observed error is

ordinary chart difficulty rather than a misleading-visual effect.

Broad multimodal evaluation suites. General suites such as MMBENCH, MME, MM-VET, LVLM-EHUB, and TINYLVLM-EHUB provide wide capability coverage but mix many failure modes into one aggregate score (Liu et al., 2024; Fu et al., 2023; Yu et al., 2024; Xu et al., 2023; Shao et al., 2023). Expert-reasoning and science/math benchmarks such as MMMU, SCIENCEQA, MATHVISTA, MATHVERSE, and multimodal chain-of-thought evaluation show that stronger reasoning scaffolds do not eliminate perception and grounding errors (Yue et al., 2024; Lu et al., 2022, 2023; Zhang et al., 2024, 2023). We use this literature mainly to justify why protocol-sensitive evaluation can add signal even when a model looks strong on broad suites.

Illusion, hallucination, and grounding diagnostics. HALLUSIONBENCH, ILLUSIONBENCH+, SEEING IS BELIEVING?, and REACTBENCH push on visually inconsistent or causally entangled evidence (Guan et al., 2024; Zhang et al., 2025; Hou et al., 2026; Zhou et al., 2026). Complementary diagnosis work such as EYES WIDE SHUT?, *Vision Language Models are Blind*, systematic hallucination audits, object-hallucination benchmarks, and multi-object hallucination analysis shows that VLM failures often reflect weak visual grounding rather than pure language misunderstanding (Tong et al., 2024; Rahmanzadehgervi et al., 2024; Seth et al., 2025; Li et al., 2023) (Chen et al., 2024; Zhou et al., 2024b). These lines motivate the external comparator side of v2, but they do not isolate the effect of changing only the response contract on matched misleading-chart items.

Misleading charts and mitigation. CHARTOM and MISLEADING-CHARTQA come closest to deceptive-chart evaluation, while recent mitigation work studies mixed misleading-visual sources rather than only deceptive charts (Bharti et al., 2025; Chen et al., 2025; Tonglet et al., 2026). Our contribution is narrower than a general mitigation or leaderboard paper: we keep a frozen misleading-chart surface and ask whether forced-answer prompting hides abstention-sensitive failures that become observable once same-item refusal is allowed. That is the specific protocol gap addressed here.

3 Method and Experimental Setup

3.1 The MMR-Trap Benchmark (v2)

The benchmark object is unchanged from v1. MMR-TRAP keeps five trap families: broken-scale, mismatched-legend, caption-conflict, proportion-illusion, and label-swap. The frozen test surface contains 30 slices of 200 examples each: five families, three severities, and both control and trap variants. This yields 6,000 scored examples per prompt regime and model. Appendix Figure 8 recaps one matched severity-2 control/trap pair per family; v2 changes the prompting and reporting protocol, not the benchmark taxonomy itself.

3.2 Dual-Prompt Evaluation

The key change in v2 is the response protocol. Every example is queried twice: once with the original forced-MCQ instruction and once with an abstain-allowed instruction that adds an explicit “insufficient evidence” option. Control items remain ordinary answerable multiple-choice questions. Trap items are deliberately constructed so that a surface cue invites a confident but unsupported answer; under the abstain-allowed regime, refusal is therefore a meaningful and often desirable behavior rather than a formatting error. Figure 2 summarizes that paired evaluation flow.

The paper-facing evaluation harness is a released MMR-TRAP letter-and-refusal parser rather than an ad hoc manual read. It extracts one capital answer label from every response, expands the choice set from A–D to A–E under MCQ+abstain, and in the abstain-enabled regime also maps a fixed refusal-marker list in free-form text to the abstain label E. Trap items are constructed so that abstention is the only correct outcome once the chart becomes internally inconsistent, while control items still score against the canonical A–D answer; when a model does answer A–D on a trap, we separately log whether that wrong letter matches the seeded lure option. Forced-MCQ runs are still allowed to show rare instruction-defying refusals, which keeps non-compliant abstention observable instead of silently folding it into ordinary error.

3.3 Experimental Protocol

Across the full paper, each condition is defined by model/backend, prompt regime, benchmark split, parser/evaluator, and decoding budget. For MMR-TRAP v2, the parser is the deterministic released letter-and-refusal scorer just described; it preserves

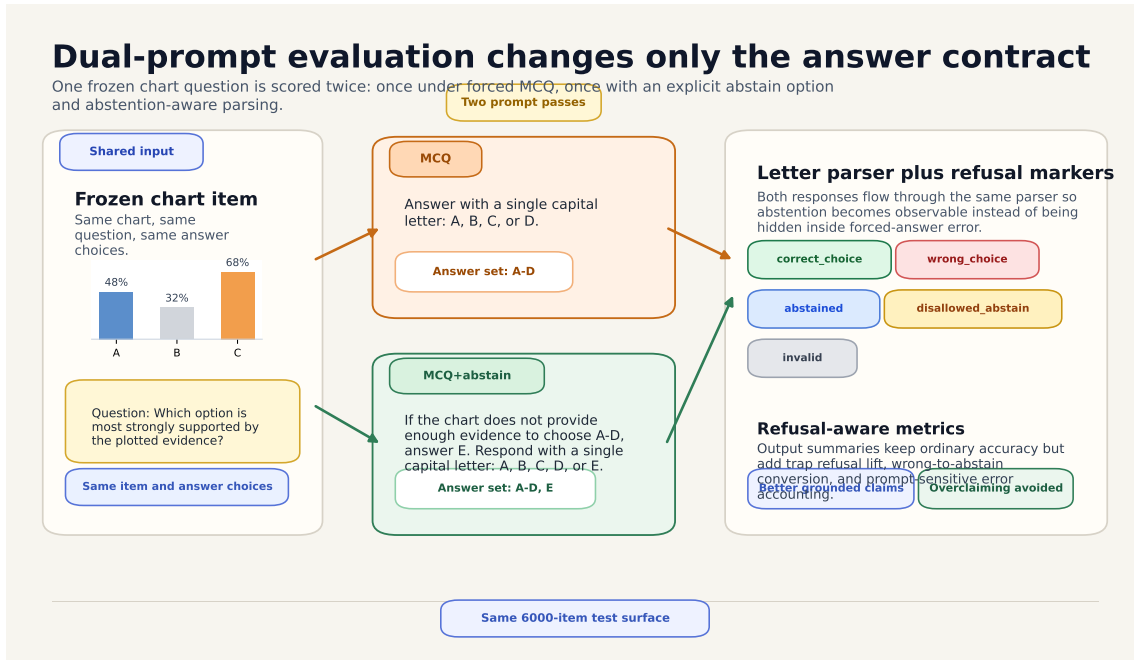


Figure 2: Dual-prompt evaluation in MMR-TRAP v2. The same frozen example is scored under a forced-answer prompt and an abstain-allowed prompt, then parsed into ordinary choice outcomes and abstention-aware outcomes so refusal becomes observable rather than structurally suppressed.

a distinct abstention outcome and a distinct trap-consistent-wrong outcome instead of collapsing both into generic error. The model slate is exactly three systems: Qwen2.5-VL-3B-Instruct and Qwen2.5-VL-7B-Instruct served through a vLLM-based inference backend, plus a provider-hosted OpenAI GPT-5 multimodal API snapshot accessed through the Responses API on 2026-06-12. All reported comparisons use one sampled response per item, random seed 0, and a 64-token output cap; the hosted OpenAI GPT-5 condition additionally fixes the provider reasoning setting to low. Reported scores always come from the benchmark-native parser or evaluator for the corresponding surface rather than from manual relabeling.

Paired MMR-TRAP setup summary. Benchmark/split: the frozen MMR-TRAP v2 test surface with 30 matched slices (five families, three severities, control plus trap) for 6,000 scored examples per prompt regime. Prompt regimes: forced MCQ versus MCQ+abstain on the same items. Parser/scoring: the released MMR-TRAP scorer extracts one answer letter, maps recognized refusal phrases to E when abstention is enabled, scores E as correct only on trap items, and records lure-matching wrong letters separately. Evaluated systems: Qwen2.5-VL-3B-Instruct, Qwen2.5-VL-7B-Instruct, and a provider-hosted OpenAI GPT-5 mul-

timodal API snapshot accessed through the Responses API on 2026-06-12. All three systems run under the same single-sample decoding policy, seed 0, and 64-token cap. Per-condition query budget: one full pass over the released split, which yields 12,000 total queries per model across the paired forced-MCQ and MCQ+abstain regimes. Under this setup the only in-domain intervention is the response protocol itself rather than benchmark composition, parser behavior, or scoring.

The same three-model slate is then reused for the focused 3×3 comparator matrix, with benchmark-native prompts and evaluators on the full released test split of each benchmark: 1,129 HallusionBench question-image tasks, 1,948 ChartQAPro items, and 1,000 ChartMuseum items. Because these benchmark families use different native metrics and mixed-quality literature anchors, we report the external matrix as benchmark-specific context rather than as a pooled ranking surface. The main evidence story is therefore intentionally restricted to one paired-prompt mechanism benchmark and one literature-anchored external comparator matrix.

External comparator setup summary. HallusionBench uses the official repository question text and its released question-accuracy evaluator on all 1,129 released question-image tasks. ChartQAPro

Field	Setting
Benchmark / source	Frozen MMR-TRAP v2 test surface inherited from the v1 five-family taxonomy.
Split / scale	30 matched slices (5 families $\times$ 3 severities $\times$ control/trap), yielding 6,000 scored examples per prompt regime.
Prompt regimes	Forced MCQ and MCQ+abstain on the same released items.
Parser / scoring	The released MMR-TRAP scorer extracts one answer letter, maps recognized refusal phrases to E when abstention is enabled, scores E as correct only on trap items, and separately records lure-matching wrong letters.
Models / backends	Qwen2.5-VL-3B-Instruct and Qwen2.5-VL-7B-Instruct via vLLM, plus OpenAI GPT-5 via a hosted Responses API backend.
Decoding / budget	One sampled response per item, seed 0, 64-token output cap, and provider low reasoning for the hosted OpenAI GPT-5 condition.
Per-model query budget	12,000 total queries across the paired forced-MCQ and MCQ+abstain regimes.

Table 1: Paired MMR-TRAP v2 setup: 6,000 scored examples per prompt regime and 12,000 paired queries per model. The only in-domain intervention is the response contract; benchmark content, parser, and decoding budget stay fixed.

uses the official Hugging Face test split and the VLMEvalKit ChartQAPro scorer on all 1,948 released test items. ChartMuseum uses the official Hugging Face test split, the released chart-question prompt, and the benchmark’s answer-equivalence judge prompt on all 1,000 test items. Every comparator condition reuses the same three model/backends, one sampled response per item, random seed 0, and a 64-token cap, but each benchmark keeps its native parser and metric. We therefore treat the external matrix as benchmark-specific context rather than as interchangeable measurements on one shared scale.

Because forced MCQ and MCQ+abstain are run on the same frozen items, every main comparison is paired by construction. We therefore use item-level exact paired tests to ask whether the answer distribution changes under the alternate response rule, not to claim seed-robust variance estimates. That distinction is important for interpretation: the paper can make a within-item mechanism claim about protocol sensitivity even though it still lacks the repeated-seed evidence needed for a stability claim.

4 Literature Anchoring

The literature-anchor audit remains per-cell rather than pooled. For each comparator row we record the closest published baseline, the delta in percentage points, and whether that anchor is within a ± 10 point tolerance band. The message is intentionally cautionary: four rows are within the tolerance band, HallusionBench Qwen2.5-VL-3B-Instruct remains unanchored because no faithful

published $\leq 4B$ Qwen-family cell exists, and the remaining four comparisons are anomalous enough to require benchmark-specific discussion. The preserved MISLEADING-CHARTQA row remains limitations-only context rather than part of the nine reported evaluations; Figure 3 plots the same per-cell deltas.

Anchor selection follows a nearest-comparable rule rather than a best-looking one. When a paper reports the exact evaluated model, we use that direct cell; when it does not, we step to the closest same-family release or the nearest closed-model predecessor and explicitly mark the comparison as indirect. If no faithful cell exists at all, as with HallusionBench for a $\leq 4B$ Qwen-family model, the row remains unanchored rather than being force-matched across architectures. The audit is therefore conservative by design: it prefers an explicit “no published baseline” status to a cleaner-looking but scientifically weaker parity claim.

5 Results

The central result is the paired-prompt evaluation. Table 2 summarizes the aggregated paired-prompt results for the three paper models.

Family-Level Redistribution. Figure 4 shows that the gains in Table 2 are driven by a few families rather than by a uniform lift. For GPT-5, trap accuracy remains effectively at floor on broken-scale, mismatched-legend, and proportion-illusion (0.7%, 0.0%, and 1.5%), and rises mainly on caption-conflict (14.8%) and label-swap (8.0%). These trap gains almost exactly match the corresponding trap-side refusal rates, which means the recovery

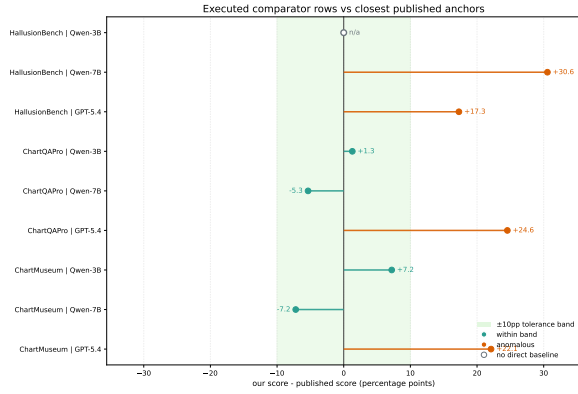


Figure 3: Executed comparator rows against the closest published baselines. Green rows fall within the ± 10 point tolerance band, the gray HallusionBench Qwen-3B row has no faithful direct baseline, and orange rows exceed the band and therefore require benchmark-specific discussion rather than parity claims.

Model	Acc MCQ	Acc +Abs	Ref MCQ	Ref +Abs	TrapRef MCQ	TrapRef +Abs
OpenAI GPT-5	50.0	52.5	0.0	2.5	0.0	5.0
Qwen2.5-VL-3B	33.6	37.6	0.0	8.9	0.0	12.0
Qwen2.5-VL-7B	46.9	61.1	1.1	23.9	1.0	37.5

Table 2: Completed paired-prompt aggregate on the paper model slate. Allowing abstention surfaces latent refusal on trap items, with the largest trap-side shift on Qwen2.5-VL-7B (1.0% $\rightarrow$ 37.5%, a +36.5 point change). Percentages are aggregated over the full 6,000-example regime for each model; the closed-model row corresponds to the provider-hosted OpenAI GPT-5 evaluation.

comes from newly expressed abstention rather than from a large increase in direct lure resolution.

Qwen2.5-VL-3B-Instruct exposes a broader but noisier uncertainty signal. Under MCQ+abstain, trap accuracy rises to 25.8% on mismatched-legend and 24.7% on label-swap, but remains low on broken-scale (1.7%), caption-conflict (2.8%), and proportion-illusion (4.8%). The same prompt induces visible control-side refusal on label-swap (20.7%) and mismatched-legend (7.8%), with corresponding control accuracies of 79.3% and 23.3%. For the 3B model, abstention is informative, but it is not yet selective: the model often backs away from difficult controls as well as deceptive traps.

Qwen2.5-VL-7B-Instruct is the clearest evidence that the protocol shift can surface latent trap detection. Broken-scale trap accuracy rises from 0.0% to 82.5%, label-swap to 42.7%, caption-conflict to 30.0%, and mismatched-legend to 20.5%, while trap-consistent wrong behavior falls from 20.0% to 0.5% on broken-scale and from 33.7% to 21.8% on mismatched-legend. The trade-

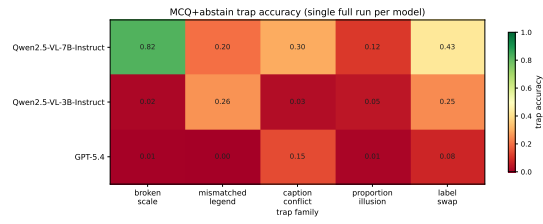


Figure 4: Trap accuracy by model and family for the abstain-allowed E5 slice. Qwen2.5-VL-7B-Instruct reaches 82.5% on broken-scale but only 11.7% on proportion-illusion, while GPT-5 stays at or below 14.8% on every trap family. The plotting scaffold retains CI slots, but the current paper reports single-run means only.

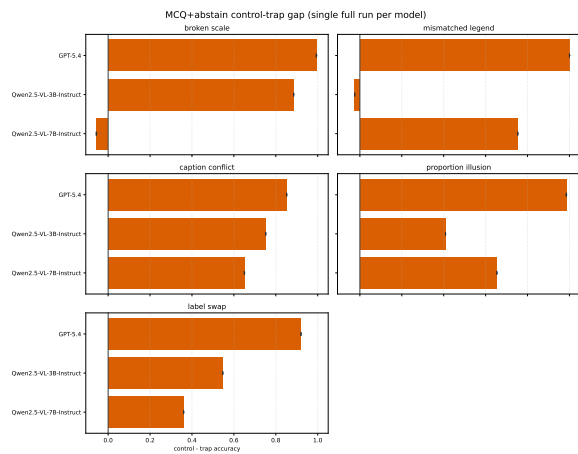


Figure 5: Under MCQ+abstain, Qwen2.5-VL-7B is the only model that flips a family-level control-trap gap negative, on broken-scale (-5.7 points), while GPT-5 stays near a full-point gap on every family. Positive values mean the matched control remains easier than the trap; negative values mean the trap recovers to or above the control.

off is over-abstention on some matched controls: broken-scale control accuracy drops from 97.5% to 76.8%, and label-swap control accuracy drops from 100.0% to 78.7%. The strongest model therefore does not simply become better at every item; it becomes more willing to withhold answers on the families where misleading structure is easiest to detect.

Selective Abstention vs. Control Cost. Figure 5 makes the same mechanism visible as a control-trap accuracy gap under MCQ+abstain rather than as a refusal count. GPT-5 keeps near-perfect control accuracy and therefore stays close to a full-point gap on every family (0.85–1.00), which means the new abstentions do little to repair the underlying trap weakness. Qwen2.5-VL-3B-Instruct

narrows that gap most on mismatched-legend, where trap accuracy slightly exceeds control accuracy (-0.025), but only because the model refuses both sides of the pair. Qwen2.5-VL-7B-Instruct is the only system that truly reverses a hard family at scale: on broken-scale, trap accuracy rises above control accuracy (-0.0567 gap) because abstention cuts trap-consistent wrong answers while preserving most of the control surface. The remaining Qwen2.5-VL-7B-Instruct gaps stay positive on caption-conflict, label-swap, mismatched-legend, and proportion-illusion (0.36 – 0.75), so the prompt change exposes uncertainty without eliminating lure sensitivity.

Analysis / Ablation. These family-specific re-distributions still yield two global patterns. First, the abstention option is not a cosmetic formatting change: it produces substantial trap-side refusal that is largely absent under forced MCQ, especially for Qwen2.5-VL-7B-Instruct. Second, the refusal increase is larger on trap items than on controls, which is what we would expect if the prompt change is surfacing uncertainty on misleading evidence rather than causing indiscriminate refusal everywhere. Exact paired tests over the shared items back both patterns: all three overall-accuracy gains and trap-side refusal increases are significant ($p < 10^{-19}$ and $p < 10^{-44}$, respectively), GPT-5 shows no control-side refusal shift ($p = 1.0$), and the trap-minus-control refusal gap remains positive for every model ($p \leq 4.9 \times 10^{-17}$). The same change also reduces trap-consistent wrong behavior, most visibly on Qwen2.5-VL-7B-Instruct, whose aggregate trap-side trap-consistent rate drops from 13.9% to 7.1%. Figure 6 compresses the aggregate shift into a model-level view, while Figures 4 and 5 show how unevenly that shift is distributed across trap families.

The paired counts show that these gains are not driven by a handful of unstable examples. For GPT-5, the +2.5 point overall gain corresponds to 150 items that become correct only under MCQ+abstain and zero items that move in the opposite direction; its trap-side refusal increase is the same 150 discordant pairs. Qwen2.5-VL-3B-Instruct is noisier but still net positive, with 479 items becoming correct only under MCQ+abstain versus 235 in reverse, and with 359 trap items newly abstained against 175 new control abstentions. Qwen2.5-VL-7B-Instruct is the clearest asymmetry: 1,159 items become correct only under

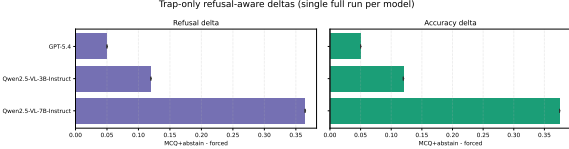


Figure 6: Allowing abstention raises trap accuracy by +5.0, +12.0, and +37.5 points for OpenAI GPT-5, Qwen2.5-VL-3B, and Qwen2.5-VL-7B, while trap-side refusal rises by +5.0, +12.0, and +36.5 points. The panel summarizes one 6,000-example paired-prompt evaluation per model, so it reports single-run aggregates rather than between-seed confidence intervals.

MCQ+abstain versus 307 in reverse, and trap-side abstention rises on 1,101 items while only 7 trap items show forced-only refusal. Those discordant counts make the mechanism claim concrete: the prompt change systematically reclassifies a large set of trap responses rather than nudging a small set of borderline cases.

Failure Cases. The recovery is concentrated rather than uniform. Qwen2.5-VL-7B-Instruct reaches 82.5% on broken-scale while GPT-5 remains below 15% on every trap family, and Appendix Figure 9 preserves that asymmetry across severities: Qwen2.5-VL-7B-Instruct stays above 76% on broken-scale across all three severities, whereas GPT-5 remains at or near zero on broken-scale, mismatched-legend, and proportion-illusion even after abstention is allowed. These family-level and severity-level slices are descriptive views from a single paired-prompt evaluation per model rather than between-seed estimates.

6 Cross-Benchmark Comparator

The cross-benchmark matrix is exactly three models by three external benchmarks: HALLUSION-BENCH, CHARTQAPRO, and CHARTMUSEUM. Table 3 is the reader-facing matrix for that evidence. It keeps benchmark-native metrics, task scale, and decoding/budget facts visible in one place rather than asking the reader to reconstruct them from figure annotations.

Chart benchmarks. Four rows fall within the literature tolerance band, and those rows matter because they show that the comparator matrix is capable of producing ordinary-looking benchmark behavior. On CHARTQAPRO, Qwen2.5-VL-3B-Instruct is only +1.3 points from the closest small-Qwen anchor and Qwen2.5-VL-7B-Instruct is -5.4 points from the closest 7B anchor. On

Benchmark / split	Tasks	Metric	Budget / decoding	OpenAI GPT-5 hosted API	Qwen2.5-VL-3B vLLM backend	Qwen2.5-VL-7B vLLM backend	Anchor status
HALLUSIONBENCH test	1,129 question-image tasks	Question acc.	One full run; single-sample decoding; 64-token cap	82.6	58.4	69.7	1 unanchored row, 2 anomalous rows
CHARTQAPRO test	1,948 items	Overall acc.	One full run; single-sample decoding; 64-token cap	62.2	22.0	30.2	2 within-band rows, 1 anomalous row
CHARTMUSEUM test	1,000 items	Overall acc.	One full run; single-sample decoding; 64-token cap	64.3	28.2	19.6	2 within-band rows, 1 anomalous row

Table 3: Main cross-benchmark results matrix for the focused 3×3 scope. Bold marks the highest benchmark-native score in each row. The table is intentionally benchmark-specific rather than pooled because the three families use different released test surfaces, different native metrics, and mixed-quality literature anchors. Figure 7 visualizes the same nine cells with anchor-status coloring.

CHARTMUSEUM, the same two models land at +7.2 and −7.2 points, respectively. These within-band rows suggest that the open-weight models behave plausibly on ordinary chart reasoning tasks once the benchmark does not force the misleading-visual mechanism under study. The closed-model rows on those same benchmarks move in the opposite direction: OpenAI GPT-5 leads the raw scores, but its deltas of +24.6 and +22.1 points are too large to interpret as straightforward parity with the closest published GPT-4o anchors.

HallusionBench and closed-model drift. HALLUSIONBENCH is the clearest reminder that external validity depends on the benchmark family and the quality of the nearest published anchor. Qwen2.5-VL-3B-Instruct reaches 58.4% question accuracy, but there is no faithful published $\leq 4B$ Qwen-family comparator, so the row is unanchored rather than parity-claimable. Qwen2.5-VL-7B-Instruct reaches 69.7%, which is 30.6 points above the closest published Qwen-family cell; that gap is too large to interpret as a clean reproduction. OpenAI GPT-5 is numerically strongest on HALLUSIONBENCH as well, but its +17.3 point delta to the nearest GPT4V-style anchor may reflect model-generation drift, evaluator differences, or release mismatch rather than a clean scientific ranking.

7 Discussion

The paired-prompt evaluation supports one local mechanism claim: on MMR-TRAP v2, part of the apparent misleading-visual failure under forced-answer evaluation is suppressed uncertainty reporting. OpenAI GPT-5 converts only a small caption-conflict and label-swap slice into abstentions, Qwen2.5-VL-3B-Instruct expresses broader uncertainty at noticeable control-side cost, and Qwen2.5-VL-7B-Instruct shows the clearest selec-

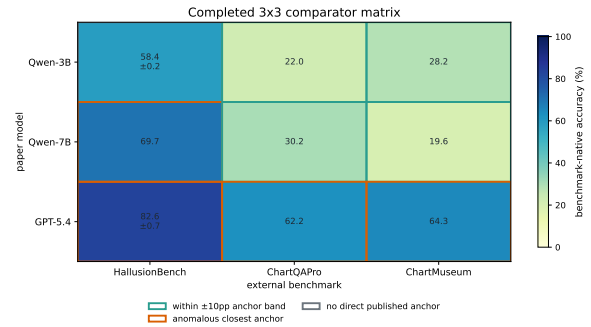


Figure 7: The focused 3×3 comparator matrix contains four within-band rows, one unanchored HallusionBench row, and four anomalous closest-anchor comparisons. Cell values show benchmark-native accuracy, while border colors encode literature-anchor status.

tive trap response, gaining 14.2 accuracy points while still increasing control-side refusal by 9.2 points. Accuracy alone therefore hides trap accuracy, control-side refusal cost, and the conversion of trap-consistent wrong answers into abstentions. The external 3×3 matrix sets the boundary on that claim: it lacks the matched-item protocol control of MMR-TRAP and still contains indirect or anomalous anchors, so we use it as scope-limiting context rather than pooled evidence of general cross-benchmark superiority.

8 Conclusion

MMR-TRAP v2 narrows the v1 benchmark into a mechanism paper about protocol-sensitive uncertainty. On this benchmark, the paired-prompt results show that a forced-answer protocol can hide substantial abstention on misleading visuals, especially for Qwen2.5-VL-7B-Instruct, and therefore change both refusal and accuracy measurements. The comparator matrix adds useful external context but also limits the claim: four rows match nearby anchors, one row is unanchored, and four remain

anomalous. The resulting thesis is intentionally local: dual prompting reveals hidden uncertainty on MMR-TRAP v2, while the mixed comparator evidence constrains how broadly that result should be generalized.

9 Limitations

Synthetic surface and no human ceiling. The core MMR-TRAP items remain programmatic misleading visuals rather than collected real-world charts or images, and the study still lacks a human upper bound; both omissions matter because a credible ceiling would require a dedicated annotation effort beyond the current scope.

Mixed external anchors. The three-benchmark comparator matrix is complete, but it is anchored only weakly to prior work: four rows are within the tolerance band, one row has no faithful direct baseline, and four closest-anchor comparisons are anomalous. These deviations may reflect model-generation drift, evaluator differences, or benchmark protocol mismatch, so the comparator evidence remains benchmark-specific context rather than a clean parity claim.

Release and variance limits. ILLUSION-BENCH+ remains related-work context only because the public release is externally incomplete, MISLEADING-CHARTQA remains a limitations note because its released evaluator and predictions do not support a like-for-like baseline reconstruction, and the paired-prompt mechanism evidence still comes from one full evaluation per model rather than a seed-robust study. The provider-hosted closed-model row is a dated OpenAI GPT-5 API snapshot, so evaluator drift and hidden back-end revision cannot be separated as cleanly as for the open-weight models. The teaser and dual-prompt protocol figures are hand-authored vector diagrams rather than generative-model renders, and they should be read as explanatory schematics rather than as evaluation outputs. We therefore report item-level paired significance, but we do not claim run-to-run stability.

Single-template abstention prompt. The paired regime isolates one concrete protocol change: the benchmark either forces an A–D answer or adds an explicit abstain option with a fixed parser contract. That design cleanly tests whether allowing refusal reveals hidden uncertainty on the same items, but it does not estimate how sensitive the recovered effect size is to alternative abstention

phrasings, looser free-form refusal channels, or parser variants that recognize a broader set of uncertainty expressions. Future work should therefore treat the current effect sizes as evidence that the refusal channel matters, not as a prompt-invariant estimate of its exact magnitude.

Ethical Considerations. The study uses automated model evaluation on synthetic misleading visuals and does not collect human annotations. We treat the benchmark as a diagnostic surface for abstention and cue-following behavior rather than as evidence about human deception or a recipe for persuasive real-world chart manipulation.

References

Shubham Bharti, Shiyun Cheng, Jihyun Rho, Jianrui Zhang, Mu Cai, Yong Jae Lee, Martina Rau, and Xiaojin Zhu. 2025. Chartom: A visual theory-of-mind benchmark for llms on misleading charts. *Preprint*, arXiv:2408.14419.

Ritwick Chaudhry, Sumit Shekhar, Utkarsh Gupta, Pranav Maneriker, Prann Bansal, and Ajay Joshi. 2019. LEAF-QA: Locate, encode & attend for figure question answering. *Preprint*, arXiv:1907.12861.

Xuwei Chen, Ziqiao Ma, Xuejun Zhang, Sihan Xu, Shengyi Qian, Jianing Yang, David F. Fouhey, and Joyce Chai. 2024. Multi-object hallucination in vision-language models. *Preprint*, arXiv:2407.06192.

Zixin Chen, Sicheng Song, Kashun Shum, Yanna Lin, Rui Sheng, Weiqi Wang, and Huamin Qu. 2025. Unmasking deceptive visuals: Benchmarking multimodal large language models on misleading chart question answering. *Preprint*, arXiv:2503.18172.

Chaoyou Fu, Peixian Chen, Yunhang Shen, Yulei Qin, Mengdan Zhang, Xu Lin, Jinrui Yang, Xiaowu Zheng, Ke Li, Xing Sun, Yunsheng Wu, Rongrong Ji, Caifeng Shan, and Ran He. 2023. MME: A comprehensive evaluation benchmark for multimodal large language models. *Preprint*, arXiv:2306.13394.

Tianrui Guan, Fuxiao Liu, Xiyang Wu, Ruiqi Xian, Zongxia Li, Xiaoyu Liu, Xijun Wang, Lichang Chen, Furong Huang, Yaser Yacoob, Dinesh Manocha, and Tianyi Zhou. 2024. Hallusionbench: An advanced diagnostic suite for entangled language hallucination and visual illusion in large vision-language models. *Preprint*, arXiv:2310.14566.

Wenjin Hou, Wei Liu, Han Hu, Xiaoxiao Sun, Serena Yeung-Levy, and Hehe Fan. 2026. Seeing is believing? a benchmark for multimodal large language models on visual illusions and anomalies. *Preprint*, arXiv:2602.01816.

677	Ronghang Hu, Amanpreet Singh, Trevor Darrell, and	Nitesh Methani, Pritha Ganguly, Mitesh M. Khapra,	733
678	Marcus Rohrbach. 2020. Iterative answer prediction	and Pratyush Kumar. 2020. PlotQA: Reasoning over	734
679	with pointer-augmented multimodal transformers for	scientific plots. <i>Preprint</i> , arXiv:1909.00997.	735
680	TextVQA. <i>Preprint</i> , arXiv:1911.06258.		
681	Mohammed Saidul Islam, Raian Rahman, Ahmed	Pooyan Rahmanzadehgervi, Logan Bolton, Moham-	736
682	Masry, Md Tahmid Rahman Laskar, Mir Tafseer	mad Reza Taesiri, and Anh Totti Nguyen. 2024. Vi-	737
683	Nayeem, and Enamul Hoque. 2024. Are large vi-	sion language models are blind. In <i>Computer Vision</i>	738
684	sion language models up to the challenge of chart	– ACCV 2024, pages 293–309.	739
685	comprehension and reasoning? an extensive investi-		
686	gation into the capabilities and limitations of LVLMs.	Ashish Seth, Dinesh Manocha, and Chirag Agarwal.	740
687	<i>Preprint</i> , arXiv:2406.00257.	2025. Towards a systematic evaluation of halluci-	741
688	Kushal Kafle, Brian Price, Scott Cohen, and Christo-	nations in large-vision language models. <i>Preprint</i> ,	742
689	pher Kanan. 2018. DVQA: Understanding data	arXiv:2412.20622.	743
690	visualizations via question answering. <i>Preprint</i> ,		
691	arXiv:1801.08163.	Wenqi Shao, Meng Lei, Yutao Hu, Peng Gao, Kaipeng	744
692	Samira Ebrahimi Kahou, Vincent Michalski, Adam	Zhang, Fanqing Meng, Peng Xu, Siyuan Huang,	745
693	Atkinson, Akos Kadar, Adam Trischler, and	Hongsheng Li, Yu Qiao, and Ping Luo. 2023.	746
694	Yoshua Bengio. 2018. FigureQA: An annotated	TinyLVLM-eHub: Towards comprehensive and ef-	747
695	figure dataset for visual reasoning. <i>Preprint</i> ,	ficient evaluation for large vision-language models.	748
696	arXiv:1710.07300.	<i>Preprint</i> , arXiv:2308.03729.	749
697	Yifan Li, Yifan Du, Kun Zhou, Jinpeng Wang,	Liyan Tang, Grace Kim, Xinyu Zhao, Thom Lake,	750
698	Wayne Xin Zhao, and Ji-Rong Wen. 2023. Eval-	Wenxuan Ding, Fangcong Yin, Prasann Singhal,	751
699	uating object hallucination in large vision-language	Manya Wadhwa, Zeyu Leo Liu, Zayne Sprague,	752
700	models. <i>Preprint</i> , arXiv:2305.10355.	Ramya Namuduri, Bodun Hu, Juan Diego Rodriguez,	753
701	Yuan Liu, Haodong Duan, Yuanhan Zhang, Bo Li,	Puyuan Peng, and Greg Durrett. 2026. Chartmuseum:	754
702	Songyang Zhang, Wangbo Zhao, Yike Yuan, Jiaqi	Testing visual reasoning capabilities of large vision-	755
703	Wang, Conghui He, Ziwei Liu, Kai Chen, and Dahua	language models. <i>Preprint</i> , arXiv:2505.13444.	756
704	Lin. 2024. MMBench: Is your multi-modal model		
705	an all-around player? <i>Preprint</i> , arXiv:2307.06281.	Shengbang Tong, Zhuang Liu, Yuexiang Zhai, Yi Ma,	757
706	Pan Lu, Hritik Bansal, Tony Xia, Jiacheng Liu, Chun-	Yann LeCun, and Saining Xie. 2024. Eyes wide	758
707	yuan Li, Hannaneh Hajishirzi, Hao Cheng, Kai-	shut? exploring the visual shortcomings of multi-	759
708	Wei Chang, Michel Galley, and Jianfeng Gao. 2023.	modal LLMs. In <i>Proceedings of the IEEE/CVF Con-</i>	760
709	MathVista: Evaluating mathematical reasoning of	ference on Computer Vision and Pattern Recognition	761
710	foundation models in visual contexts. <i>Preprint</i> ,	, pages 9568–9578.	762
711	arXiv:2310.02255.	Jonathan Tonglet, Tinne Tuytelaars, Marie-Francine	763
712	Pan Lu, Swaroop Mishra, Tony Xia, Liang Qiu, Kai-	Moens, and Iryna Gurevych. 2026. Protecting mul-	764
713	Wei Chang, Song-Chun Zhu, Øyvind Tafjord, Peter	timodal large language models against misleading	765
714	Clark, and Ashwin Kalyan. 2022. Learn to explain:	visualizations. <i>Preprint</i> , arXiv:2502.20503.	766
715	Multimodal reasoning via thought chains for science		
716	question answering. <i>Preprint</i> , arXiv:2209.09513.	Peng Xu, Wenqi Shao, Kaipeng Zhang, Peng Gao,	767
717	Ahmed Masry, Mohammed Saidul Islam, Mahir Ahmed,	Shuo Liu, Meng Lei, Fanqing Meng, Siyuan Huang,	768
718	Aayush Bajaj, Firoz Kabir, Aaryaman Kartha,	Yu Qiao, and Ping Luo. 2023. LVLM-eHub: A com-	769
719	Md Tahmid Rahman Laskar, Mizanur Rahman,	prehensive evaluation benchmark for large vision-	770
720	Shadikur Rahman, Mehrad Shahmohammadi, Megh	language models. <i>Preprint</i> , arXiv:2306.09265.	771
721	Thakkar, Md Rizwan Parvez, Enamul Hoque, and		
722	Shafiq Joty. 2025. Chartqapro: A more diverse and	Weihaoyu, Zhengyuan Yang, Linjie Li, Jianfeng Wang,	772
723	challenging benchmark for chart question answering.	Kevin Lin, Zicheng Liu, Xinchao Wang, and Li-	773
724	<i>Preprint</i> , arXiv:2504.05506.	juan Wang. 2024. MM-Vet: Evaluating large multi-	774
725	Ahmed Masry, Do Xuan Long, Jia Qing Tan, Shafiq Joty,	modal models for integrated capabilities. <i>Preprint</i> ,	775
726	and Enamul Hoque. 2022. Chartqa: A benchmark	arXiv:2308.02490.	776
727	for question answering about charts with visual and	Xiang Yue, Yuansheng Ni, Kai Zhang, Tianyu Zheng,	777
728	logical reasoning. <i>Preprint</i> , arXiv:2203.10244.	Ruoqi Liu, Ge Zhang, Samuel Stevens, Dongfu	778
729	Minesh Mathew, Viraj Bagal, Rubén Pérez Tito, Di-	Jiang, Weiming Ren, Yuxuan Sun, Cong Wei, Botao	779
730	mosthenis Karatzas, Ernest Valveny, and C. V.	Yu, Ruibin Yuan, Renliang Sun, Ming Yin, Boyuan	780
731	Jawahar. 2021. InfographicVQA. <i>Preprint</i> ,	Zheng, Zhenzhu Yang, Yibo Liu, Wenhao Huang, and	781
732	arXiv:2104.12756.	3 others. 2024. Mmmu: A massive multi-discipline	782
		multimodal understanding and reasoning benchmark	783
		for expert agi. <i>Preprint</i> , arXiv:2311.16502.	784
		Renrui Zhang, Dongzhi Jiang, Yichi Zhang, Haokun	785
		Lin, Ziyu Guo, Pengshuo Qiu, Aojun Zhou, Pan Lu,	786
		Kai-Wei Chang, Peng Gao, and Hongsheng Li. 2024.	787
		MathVerse: Does your multi-modal LLM truly see	788

the diagrams in visual math problems? *Preprint*, arXiv:2403.14624.

Yiming Zhang, Zicheng Zhang, Xinyi Wei, Xiaohong Liu, Guangtao Zhai, and Xiongkuo Min. 2025. Illusionbench+: A large-scale and comprehensive benchmark for visual illusion understanding in vision-language models. *Preprint*, arXiv:2501.00848.

Zhuosheng Zhang, Aston Zhang, Mu Li, Hai Zhao, George Karypis, and Alex Smola. 2023. Multi-modal chain-of-thought reasoning in language models. *Preprint*, arXiv:2302.00923.

Shizhe Zhou, Bohan Jia, Kai Wu, Yan Shen, Tongyun Li, Yuyang Wu, and Shaohui Lin. 2026. React-bench: A cause-driven benchmark for multimodal hallucination via systematic evaluation. *Preprint*, arXiv:2605.29579.

Wei Zhou, Mohsen Mesgar, Heike Adel, and Annemarie Friedrich. 2024a. FREB-TQA: A fine-grained robustness evaluation benchmark for table question answering. *Preprint*, arXiv:2404.18585.

Yiyang Zhou, Chenhang Cui, Jaehong Yoon, Linjun Zhang, Zhun Deng, Chelsea Finn, Mohit Bansal, and Huaxiu Yao. 2024b. Analyzing and mitigating object hallucination in large vision-language models. *Preprint*, arXiv:2310.00754.

Zifeng Zhu, Mengzhao Jia, Zhihan Zhang, Lang Li, and Meng Jiang. 2025. Multichartqa: Benchmarking vision-language models on multi-chart problems. *Preprint*, arXiv:2410.14179.

A Reproducibility Appendix

The reported study evaluates the three-model slate on the frozen MMR-TRAP v2 test surface under forced-MCQ and MCQ+abstain prompts, and on HALLUSIONBENCH, CHARTQAPRO, and CHART-MUSEUM under each benchmark’s native metric.

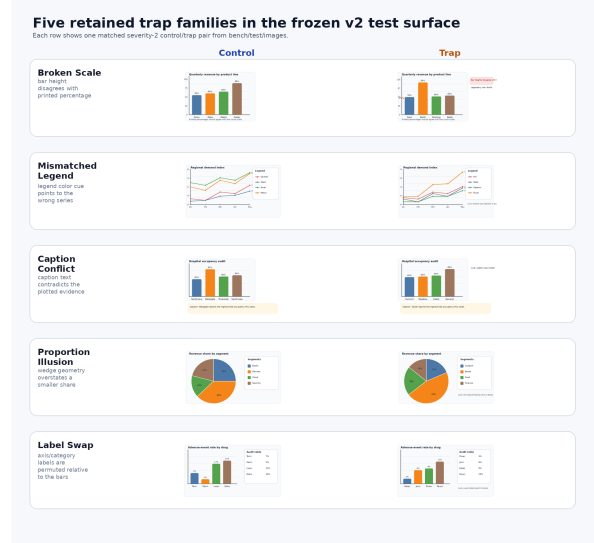


Figure 8: The five-family MMR-Trap taxonomy is inherited verbatim from v1. The recap shows one matched severity-2 control/trap pair per family from the frozen v2 test surface; this paper changes the prompting and reporting protocol, not the benchmark taxonomy itself.

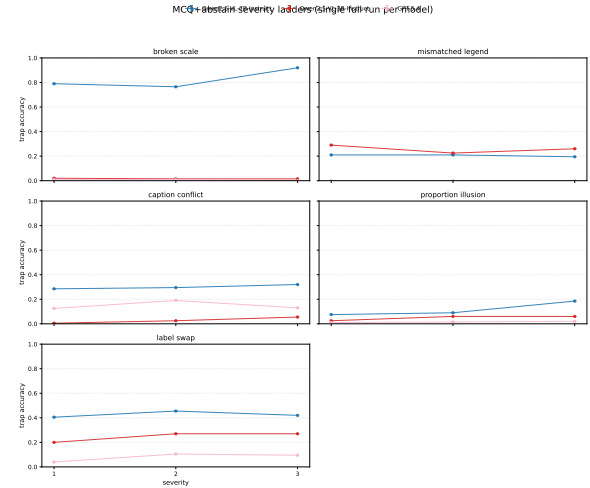


Figure 9: Per-severity trap accuracy under the abstain-allowed E5 slice. Qwen2.5-VL-7B stays above 76.5% on broken-scale across severities, whereas GPT-5 remains between 0.0% and 2.0% on broken-scale and proportion-illusion. The figure preserves the CI-ready severity layout, but the current manuscript still reports one full run per model.